

OPERATING MANUAL

MAZATROL SmoothAi
MAZATROL SmoothG
MAZATROL SmoothG³

Field Network Function
CC-Link IE Field Basic

Manual No.: H747SA0032E

Serial No.:

Before using this machine and equipment, fully understand the contents of this manual to ensure proper operation. Should any questions arise, please ask the nearest Technical Center or Technology Center.

IMPORTANT NOTICE

1. Be sure to observe the safety precautions described in this manual and the contents of the safety plates on the machine and equipment. Failure may cause serious personal injury or material damage. Please replace any missing safety plates as soon as possible.
2. No modifications are to be performed that will affect operation safety.
3. For the purpose of explaining the operation of the machine and equipment, some illustrations may not include safety features such as covers, doors, etc. Before operation, make sure all such items are in place.
4. If there is a difference between the contents of the manual and the actual state of the machine, please contact the nearest Technical Center or Technology Center.
5. Always keep this manual near the machinery for immediate use.
6. If a new manual is required, please order from the nearest Technical Center or Technology Center with the manual No. or the machine name, serial No. and manual name.
7. The export of machining programs may be restricted to prevent them from being used for military purposes. The machining programs you prepared are your property, so you should manage them yourself in accordance with the law.
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1 INTRODUCTION

A field network (fieldbus) generally refers to a means of sending data from an automation system to other systems in the vicinity of a factory site, using digital communication technologies. Field networks are divided by application into various types, such as for FA (Factory Automation), PA (Process Automation), motion control and safety, with their various communication standards.

A typical application example of the field network is to use a machine status monitoring system in order to reduce the machining time, to improve the machining quality and to keep the machine in good condition.

This NC unit supports the CC-Link IE Field Basic functions.

This manual describes the setup procedures for the CC-Link IE Field Basic function with the precautions concerned.

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1

INTRODUCTION



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2 SAFETY PRECAUTIONS

This manual is prepared only for the function or the device which is specified in the title of this manual. Therefore, you are required to carefully read and understand the operating manual of the machine, especially the section describing safety precautions, before using or operating the function or the device.

2-1 Rules

The meanings of our safety precautions to DANGER, WARNING, and CAUTION are as follows:



DANGER

: Failure to follow these instructions could result in loss of life.



WARNING

: Failure to observe these instructions could result in serious harm to a human life or body.



CAUTION

: Failure to observe these instructions could result in minor injuries or serious machine damage.

2-2 Safety Precautions Relating to the Function/Device Described in This Manual



CAUTION

- When configuring a network, push the emergency stop button for safety's sake to prevent the machine operation.
- After system configuration, check under safe conditions the data communication for correctness before operating the machine. Failure to do so could result in personal injury or machine damage.

2

SAFETY PRECAUTIONS



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3 CC-LINK IE FIELD BASIC FUNCTION

3-1 Communication Method

3-1-1 CC-Link IE Field Basic communication method

CC-Link (Control & Communication Link) IE Field Basic system is an FA network that utilizes general-purpose Ethernet. The link devices are used to perform cyclic transmission between the master station and slave stations.

In the CC-Link IE Field Basic network of the NC unit, up to 4 slave stations can be connected to the master station, when the NC unit operates as a master station. When the NC unit is used as a slave station, the number of slave stations that can be connected depends on the specifications of the master station.

3-1-2 Names and abbreviations

The names and abbreviations of CC-Link IE Field Basic-related stations, devices and data are listed in the table below.

Name and abbreviation	Description
Master station	The station that controls the entire system Only one master station exists for each network.
Slave station	A slave station is a station that cyclically transmits data to the master station. It transfers the input/output signal in units of bits, and also the input/output data in units of words.
Control master station	This is a master station which controls a slave station (as seen from a slave station).
Reservation station	This station is not actually connected, but is included in the number of network stations as a station to be connected in the future.
Cyclic transmission	This is a function which periodically performs data communication between stations in the network, using link devices.
Subnet mask	This is a number intended to facilitate control of a network to which a large number of devices are connected, by logical division in units of multiple sub-networks. The following networks constructed using Ethernet exist. • A small-scale network system in which multiple devices are connected to one Ethernet • A medium-scale or large-scale network system in which multiple connections are made using a router, for example
Disconnection	This is a type of processing that disconnects the data link when a defect occurs.
Return	This is a type of processing that re-establishes the data link when a defective station becomes normal.
Link scan (Link scanning time)	The master station sends data to the slave stations. This is the time from when a response is received from all of the slave stations, until the next request is sent.
Link device	This is a device (RX, RY, RWr, RWw) intended for communication with a slave station.
User device	This is a device (ZR) possessed by the PLC of an NC unit.
Link refresh	This function automatically transmits data between a user device and a link device.
Word	This is a unit which indicates 16-bit data. When one word is signed, it is indicated over the range of "–32768 to 32767" in decimal notation, and when it is unsigned, it is indicated over the range of "0 to 65535" in decimal notation, or over the range of "0x0000 to 0xFFFF" in hexadecimal notation.
RAS	This is an abbreviation of Reliability, Availability, and Serviceability. It refers to the overall ease of use of an automated facility.
RX	Remote input Information that is input from a slave station to the master station in units of bits. Abbreviated as RX for convenience.

Name and abbreviation	Description
RY	Remote output Information that is output from the master station to a slave station in units of bits. Abbreviated as RY for convenience.
RWw	Remote register (writing area) Information that is output from the master station to a slave station in units of 16 bits. Abbreviated as RWw for convenience.
RWr	Remote register (reading area) Information that is input from a slave station to the master station in units of 16 bits. Abbreviated as RWr for convenience.
SLMP	This is an abbreviation for Seamless Message Protocol. A protocol for accessing SLMP compatible devices from external devices.

3-1-3 Performance specifications of the CC-Link IE Field Basic function

The performance specifications of the CC-Link IE Field Basic function of the NC unit are listed in the table below.

CC-Link IE Field Basic performance specifications				
Max. number of link points per system	Remote input/output (RX, RY)		1024 points each (128 bytes)	
	Remote register (RWw) (master to slave) Remote register (RWr) (slave to master)		512 points (1024 bytes)	
Number of link points per station	Master station	Remote input/output (RX, RY)	1024 points each (128 bytes)	
		Remote register (RWw) (master to slave) Remote register (RWr) (slave to master)	512 points (1024 bytes)	
	Slave station	Remote input/output (RX, RY)	1 station occupied	64 points (8 bytes)
			2 stations occupied	128 points (16 bytes)
			3 stations occupied	192 points (24 bytes)
			4 stations occupied	256 points (32 bytes)
Slave station	Remote register (RWw) (master to slave) Remote register (RWr) (slave to master)	1 station occupied	32 points (64 bytes)	
		2 stations occupied	64 points (128 bytes)	
		3 stations occupied	96 points (192 bytes)	
		4 stations occupied	128 points (256 bytes)	
UDP port No. used in cyclic transmission		61450		
Number of steps connected in cascade		In order to verify the number of steps that can be connected when using a switching hub, contact the manufacturer of the switching hub to be used.		
Transmission path format		Star type		
Maximum distance between stations		100 m (Conforms to ANSI/TIA/EIA-568-B (Category 5e).)		
Master station reference request time		14.2 ms		
Slave station reference response time		14.2 ms		

3-1-4 Table of the CC-Link IE Field Basic functions

Among the CC-Link IE Field Basic functions, the following can be used with the NC unit.

1. Master station functions

(○: Supported ×: Not supported)

	CC-Link IE Field Basic functions		Usable/ Not usable
Cyclic transmission	Data exchange using a link device (RX/Ry/RWw/RWw)	Uses a link device (RX/Ry/RWw/RWw) to enable data to be exchanged periodically between the master station and a slave station.	○
	Link refresh	Automatically transfers data between a device in the master station and a link device.	○
	Safeguarding cyclic data	Safeguards cyclic data in units of stations.	○
	Group No. setting	By setting group numbers, the slave stations are divided into groups, and data is cyclically transferred for each group at a time.	×
	Condition of the input and output in the event of a defect	Clears or holds the condition of the input from the station containing a defective data link, and also the condition of the output that is cyclically transmitted when a stoppage error occurs in the CPU unit.	○
RAS function	Separation of the slave stations	Disconnects the applicable slave stations if there is no response within the timeout time or the number of cycles set by the slave station disconnection setting.	○
	Automatic return function	Automatically returns a disconnected station to the network and re-establishes the data link.	○
Function diagnosis	CC-Link IE Field Basic diagnosis	Enables the condition of CC-Link IE Field Basic to be verified using GxWorks2/Works3.	×
	CC-Link IE Field Basic diagnosis 2	Enables the condition of CC-Link IE Field Basic to be verified using a special relay (SM)/special register (SD).	○
Others	Reservation station setting	When a reservation station is designated, this function enables the station to be reserved for future use (although not actually connected it is included in the number of network stations as a station to be connected in the future). Although a reservation station is not actually connected, it will not be detected faultily as a defective station.	○
	Automatic detection of slave station	Under instructions from GXWorks2/3, a slave station connected to a network is detected using a Node Search command.	×

2. Slave station functions

(O: Supported ×: Not supported)

CC-Link IE Field Basic functions		Usable/ Not usable
Master conciliation function	A function which checks whether or not the master is duplicated prior to cyclic transmission, and stops the subsequent cyclic transmission if duplication occurs	○
Cyclic transmission function	A function which periodically exchanges data between master and slave stations	○
Cyclic transmission stoppage instruction function	A function whereby the master station issues instructions to a slave station to stop cyclic transmission	○
Control master station duplication detection function	A function which issues notification of the fact that when a slave station under a control master receives a cyclic transmission from another master station, the control master already exists	○
Function which detects a disconnection at a slave station	A function which detects the fact that the master station became disconnected at the slave station side	○
Group cyclic transmission function	A function that performs cyclic transmission for each group	○
Function which detects duplication of the slave station ID	A function that detects duplication of the slave station ID	○
Function which requests disconnection from the slave station	A function in which a slave station issues a disconnection request to the master station	×
Automated detection function	A function that detects a node connected to a network	○
IP address setting function	A function that sets the IP address at a node in a network	×
Parameter setting function	A function that uses an SLMP command to set the parameters of the device.	×

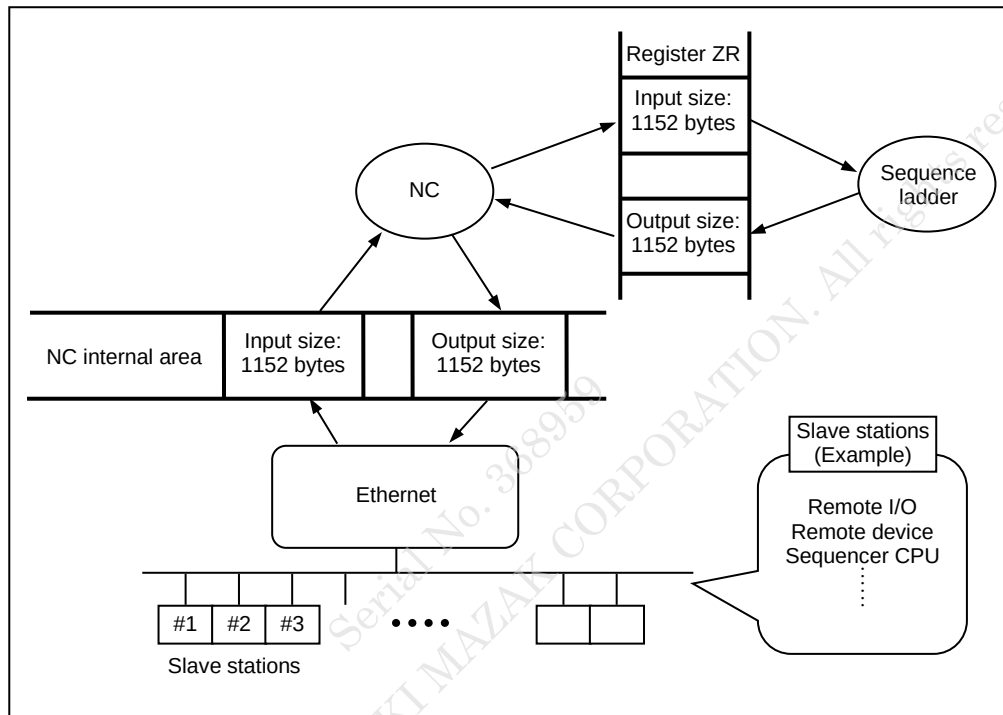
3-2 Data Input and Output

When all the CC-Link IE Field Basic communication settings are completed in the NC unit, data can be input and output between a sequence ladder and the NC unit, using a register ZR as an interface. The NC unit inputs and outputs data to and from the master station or a slave station over a CC-Link IE Field Basic network.



- For an explanation of how to make settings for CC-Link IE Field Basic communications, see “3-3 Configuring a Network.”

The overall flow of data input/output is shown in the figure below.



3-2-1 Data input/output areas

The registers ZR used as interfaces between the NC unit and a sequence ladder are listed in the table below.

RX, RY, RWr and RWw data can be freely assigned in the following areas, on condition that they are assigned in such a way that their data areas do not overlap each other.

If there are overlapping data areas, data may not be input or output normally. There is no alarm provided for warning of overlap.

Register ZR	Meaning	Corresponding remote input/output	Corresponding remote register
ZR3400 to ZR3975	Field network input data	RX	RWr
ZR5300 to ZR5875	Field network output data	RY	RWw

3-3 Configuring a Network

3-3-1 Parameter setting

In order for the NC unit to establish communication as the master station or a slave station over the CC-Link IE Field Basic network, there is a need to set parameters.

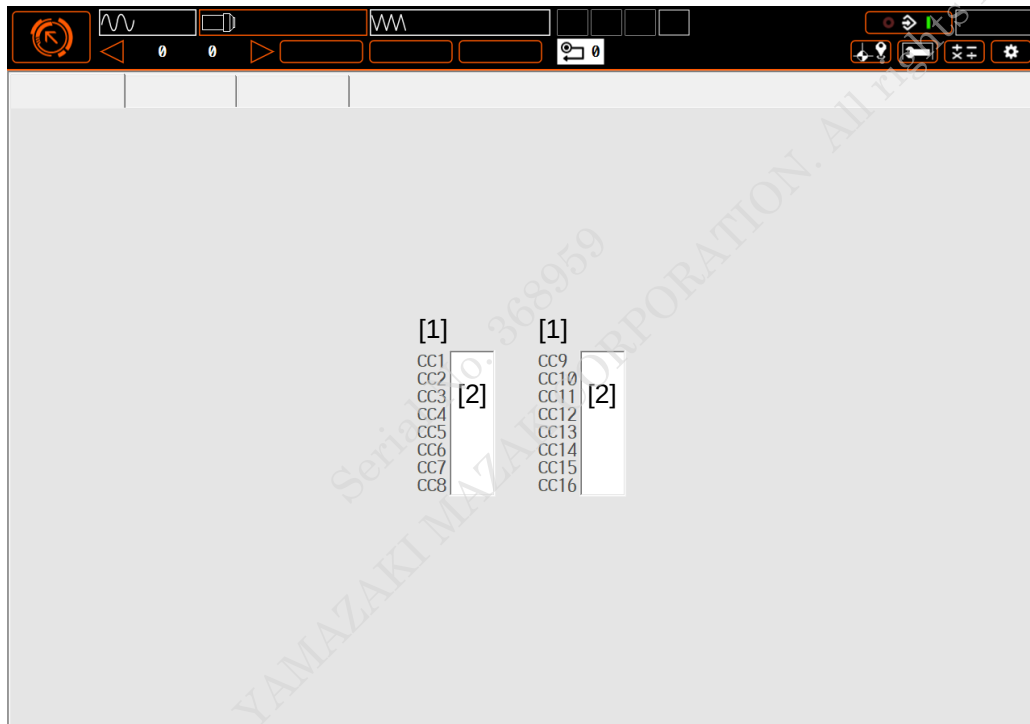
The **CC-LINK IEF BASIC PARAMETER** display is used to set parameters.

1. Display for common setting parameters of CC-Link IE Field Basic communications

Function	Allows you to register and display the parameters that need to be set for CC-Link IE Field Basic communications of the NC unit.
-----------------	---

A. Display items

Displayed data



Contents of displayed data

No.	Item	Unit	Contents of data
[1]	Address	—	Addresses of CC-Link IE Field Basic parameters for common settings
[2]	Data	—	Data of CC-Link IE Field Basic parameters for common settings

Note 1: For details, see Section “4-1 CC-Link IE Field Basic Parameters” or the separately supplied “PARAMETER LIST/ALARM LIST/M-CODE LIST.”

Note 2: Before changing a parameter, be sure to stop the PLC built into the NC unit and slave stations and check the contents of the parameter carefully.

Note 3: The parameters need to be set regardless of what role the NC unit plays: master station or slave station.

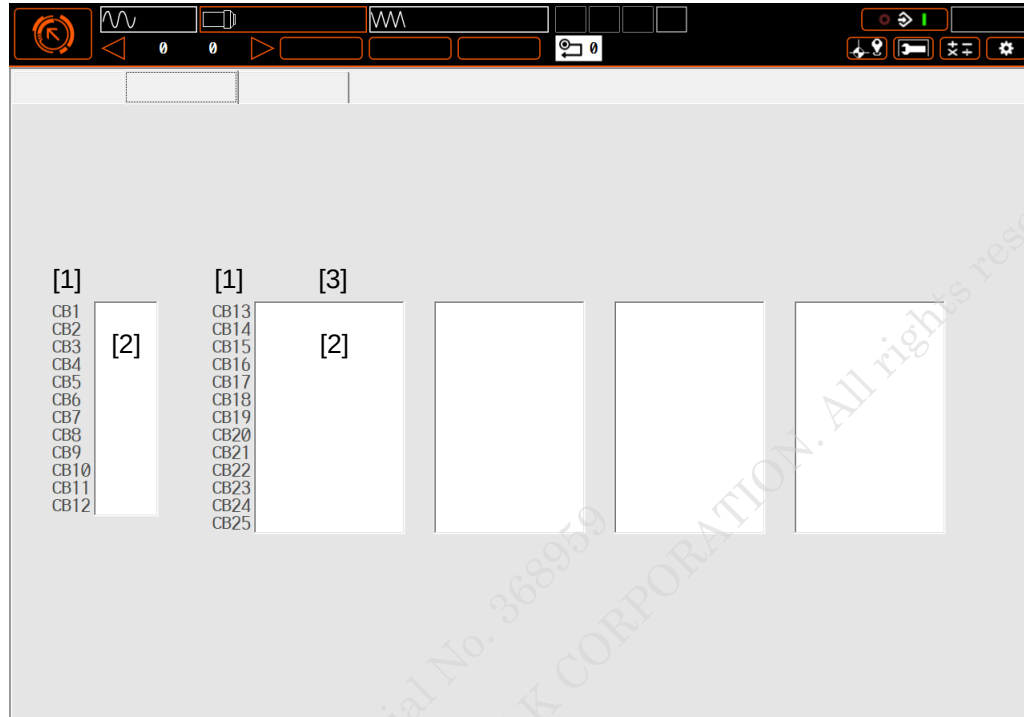
Note 4: After setting all the parameters, turn the power back on.

2. Display for master function setting parameters of CC-Link IE Field Basic communications

Function	When the NC unit is used as the master station, this display allows you to register and display the parameters that need to be set for CC-Link IE Field Basic communications.
-----------------	---

A. Display items

Displayed data



Contents of displayed data

No.	Item	Unit	Contents of data
[1]	Address	—	Addresses of CC-Link IE Field Basic parameters for master function settings
[2]	Data	—	Data of CC-Link IE Field Basic parameters for master function settings
[3]	Station number	—	Order in which slave stations are connected

Note 1: For details, see Section “4-1 CC-Link IE Field Basic Parameters” or the separately supplied “PARAMETER LIST/ALARM LIST/M-CODE LIST.”

Note 2: Before changing a parameter, check its content carefully.

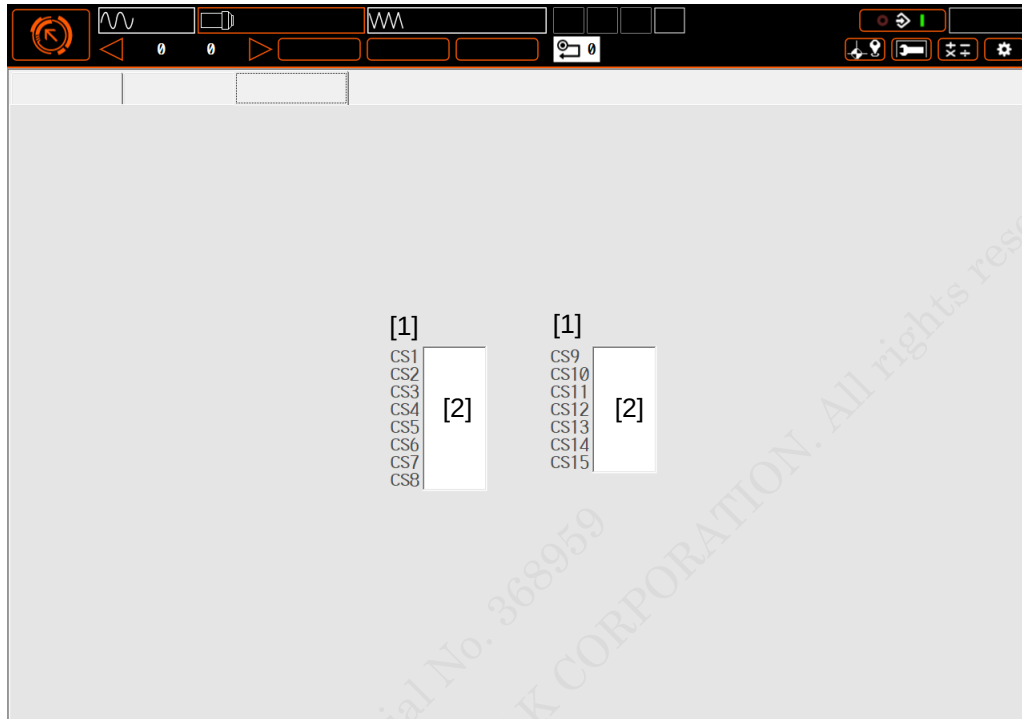
Note 3: Parameters **CB13** to **CB25** are set for the slave stations connected in the same system as the master station set by **CC2**. They need to be set in the same order as the slave stations are connected.

3. Display for slave function setting parameters of CC-Link IE Field Basic communications

Function	When the NC unit is used as a slave station, this display allows you to register and display the parameters that need to be set for CC-Link IE Field Basic communications.
-----------------	--

A. Display items

Displayed data



Contents of displayed data

No.	Item	Unit	Contents of data
[1]	Address	—	Addresses of CC-Link IE Field Basic parameters for slave function settings
[2]	Data	—	Data of CC-Link IE Field Basic parameters for slave function settings

Note 1: For details, see Section “4-1 CC-Link IE Field Basic Parameters” or the separately supplied “PARAMETER LIST/ALARM LIST/M-CODE LIST.”

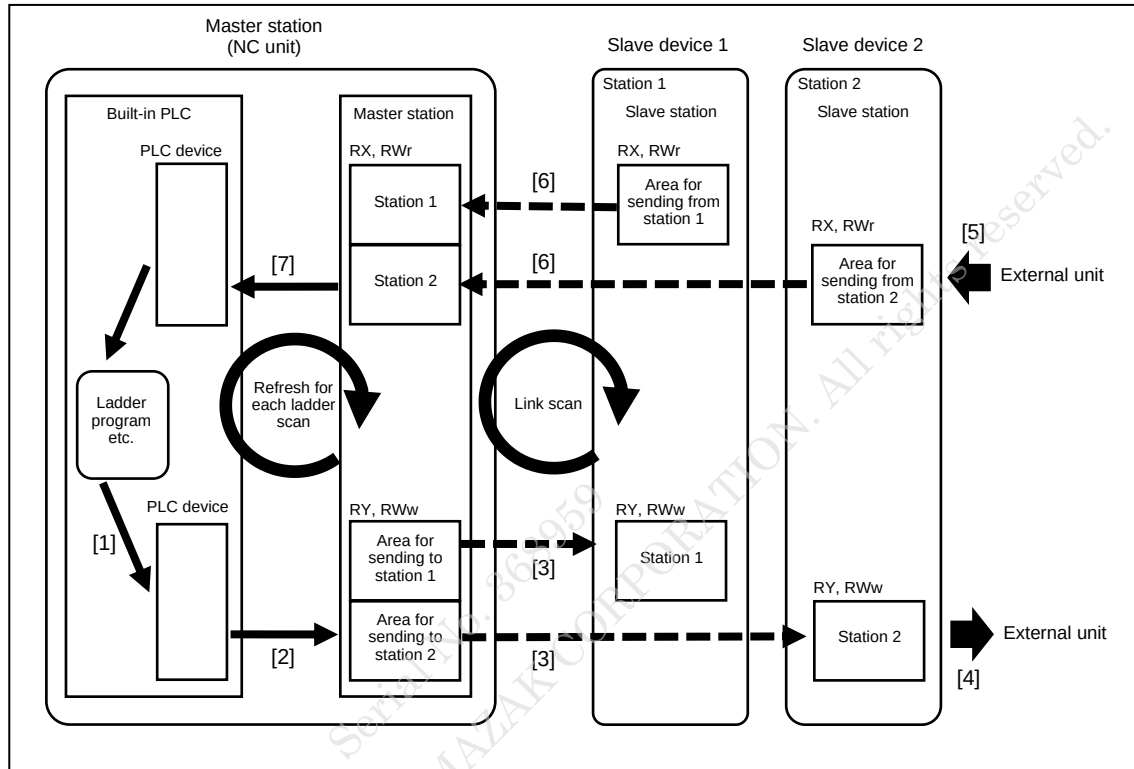
Note 2: Before changing a parameter, check its content carefully.

3-3-2 Cyclic transmission

Cyclic transmission refers to a means of transmitting bit information (remote input/output) and word information (remote register) cyclically between stations connected to the CC-Link IE Field Basic network.

1. Flow of transmitted data with the NC unit set as master station

An example of the flow of data transmitted by cyclic transmission is shown below.



A. Data transfer from the master station to a slave station

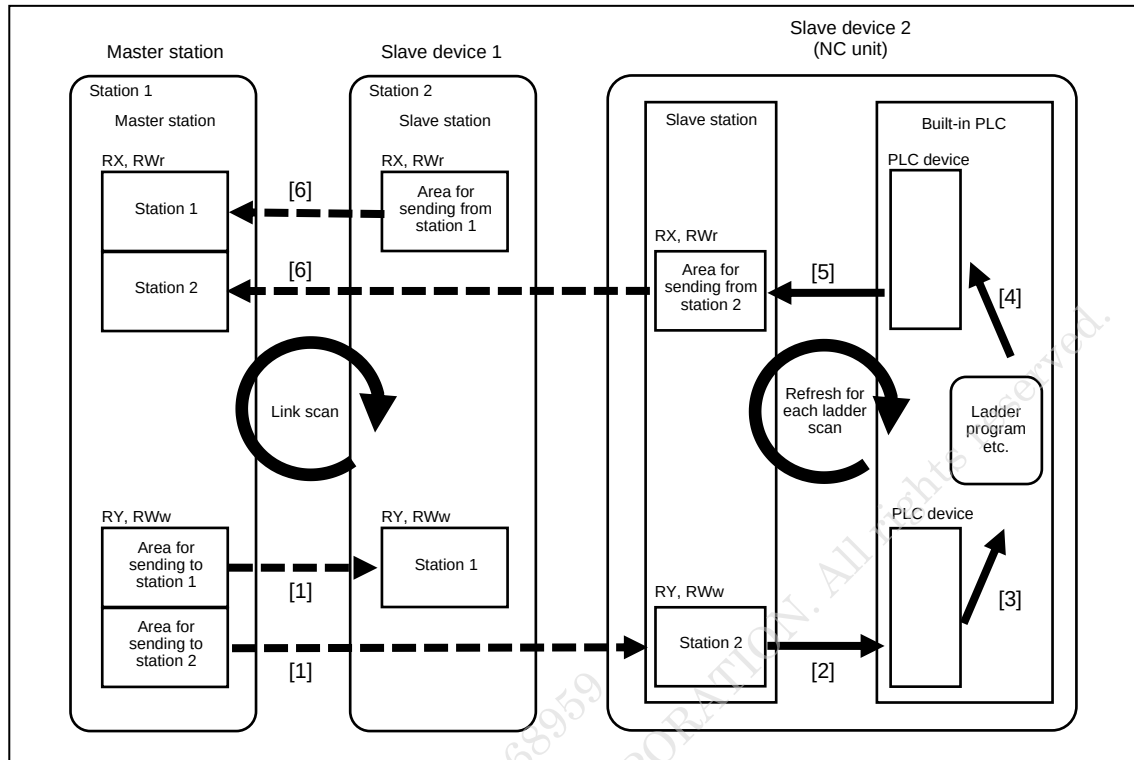
- [1] The PLC device is turned ON using a ladder program, or the like.
- [2] As a result of the refresh operation, the condition of the PLC device will be stored in link devices (RY and RWw) in the master station.
- [3] The condition of the link devices (RY and RWw) in the master station will be stored in the link devices in the slave station (RY and RWw) by means of the link scan operation.
- [4] The condition of the link devices (RY and RWw) in the slave station is output to the external unit.

B. Data transfer from a slave station to the master station

- [5] The condition of the external unit is stored in the link devices (RX and RWr) in the slave station.
- [6] The condition of the link devices (RX and RWr) in the slave station is stored in the link devices (RX and RWr) in the master station by means of the link scan operation.
- [7] The condition of the link devices (RX and RWr) in the master station is stored in the PLC device by means of the refresh operation.

2. Flow of transmitted data with the NC unit set as slave station

An example of the flow of data transmitted by cyclic transmission is shown below.



A. Data transfer from the master station to a slave station

- [1] The condition of the link devices (RY and RWw) in the master station is stored in the link devices (RY and RWw) in the slave station by means of a link scan.
- [2] The condition of the link devices (RY and RWw) in the slave station is stored in the PLC device by means of the refresh operation.
- [3] The value stored in the PLC device is referred to by means of a ladder program, or the like.

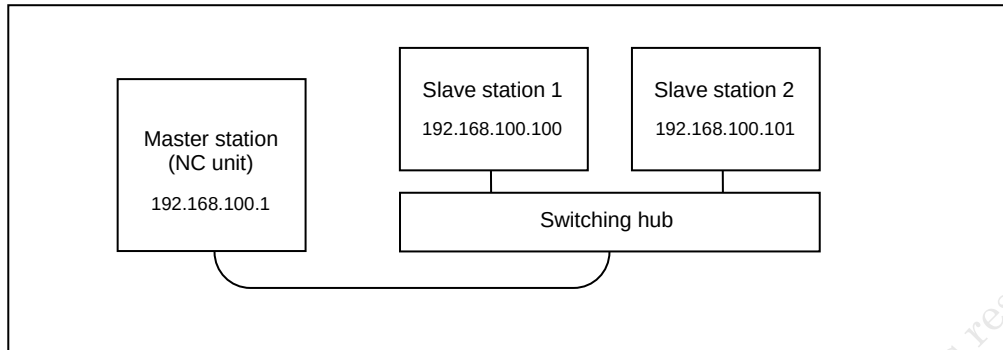
B. Data transfer from the slave station to the master station

- [4] The PLC device is turned ON using a ladder program, or the like.
- [5] The condition of the PLC device is stored in the link device (RX and RWr) in the slave station by means of the refresh operation.
- [6] The condition of the link devices (RX and RWr) in the slave station is stored in the link devices (RX and RWr) of the master station by means of a link scan.

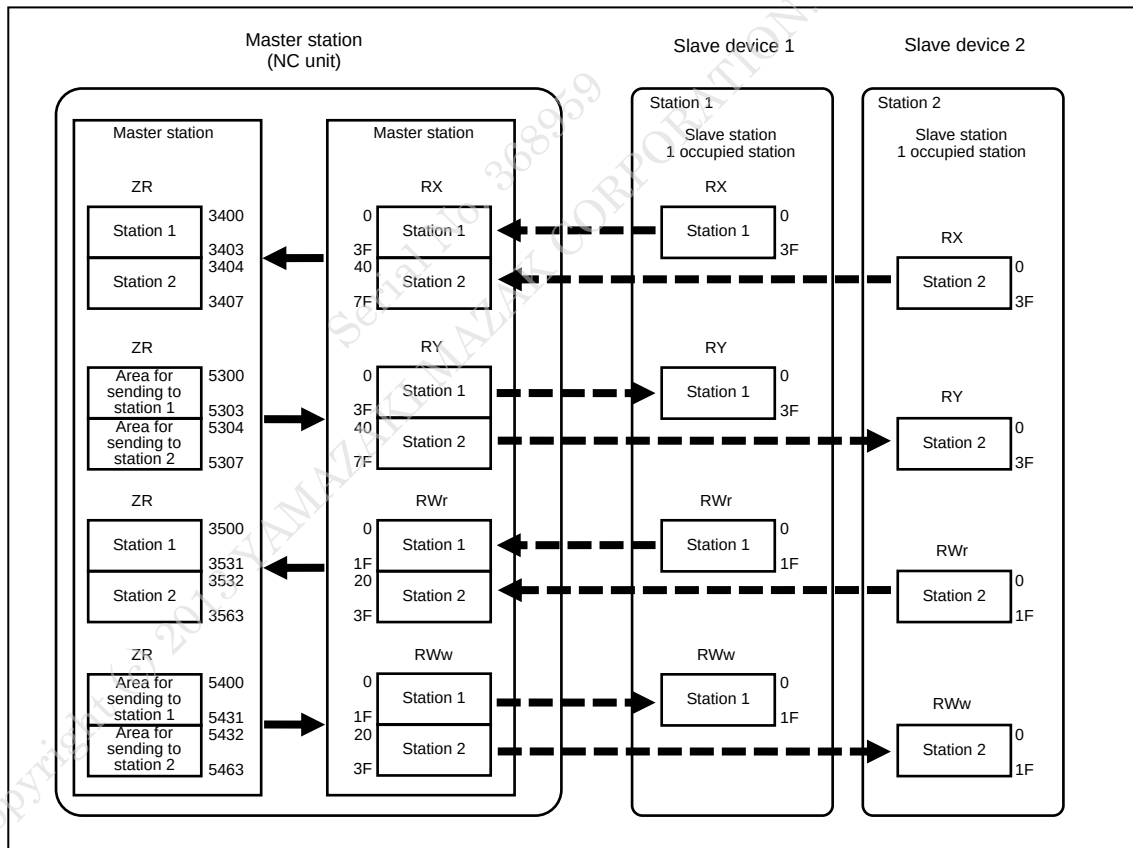
3. Examples of parameter settings on the CC-LINK IEF BASIC PARAMETER display

A. Example of parameter settings with the NC unit set as master station

Here are some examples of parameter settings when the NC unit serves as the master station and two slave stations are connected to it.



Shown below are examples in which link devices RX and RY per station are assigned at 64 points, and link devices RWr and RWw per station are assigned at 32 points, based on the assumption that there is only one occupied station in each slave device.



< Examples of settings of the parameters >

1. Enter the operation settings, for example, station type, in parameters **CC1** to **CC5** and parameters **CB1** to **CB7**.

Address	Description	Setting
CC1	Communication enabled	1: Enabled
CC2	Setting the type of station	0: Master station
CC3	Error switchover	0: Reset stoppage
CC4	Data link defective station setting	0: Clear
CC5	Output setting during PLC STOP	0: Output
CB1	Master function Number of connected devices	2
CB3	Master function Timeout time	100 (ms)
CB7	Master function Timeout time	3

2. Enter the settings in parameters **CB13** to **CB25** of the first station, for the first slave device.

Address	Description	Setting
CB13	Number of occupied stations of slave device	1
CB15	Reservation station setting of slave device	0: No setting
CB16	Refresh device project number of slave device	1
CB17	Slave device IP address	192.168.100.100
CB22	Leading refresh device number (RX) of slave device	3400
CB23	Leading refresh device number (RY) of slave device	5300
CB24	Leading refresh device number (RW _r) of slave device	3500
CB25	Leading refresh device number (RW _w) of slave device	5400

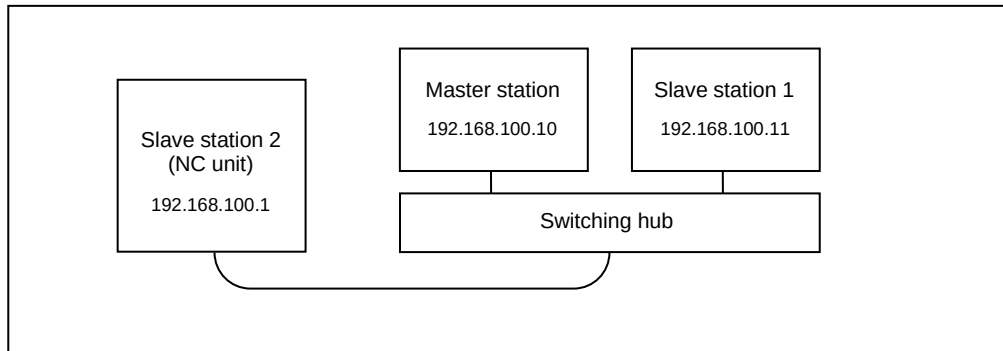
3. Enter the settings in parameters **CB13** to **CB25** of the second station, for the second slave device.

Address	Description	Setting
CB13	Number of occupied stations of slave device	1
CB15	Reservation station setting of slave device	0: No setting
CB16	Refresh device project number of slave device	1
CB17	Slave device IP address	192.168.100.101
CB22	Leading refresh device number (RX) of slave device	3404
CB23	Leading refresh device number (RY) of slave device	5304
CB24	Leading refresh device number (RW _r) of slave device	3532
CB25	Leading refresh device number (RW _w) of slave device	5432

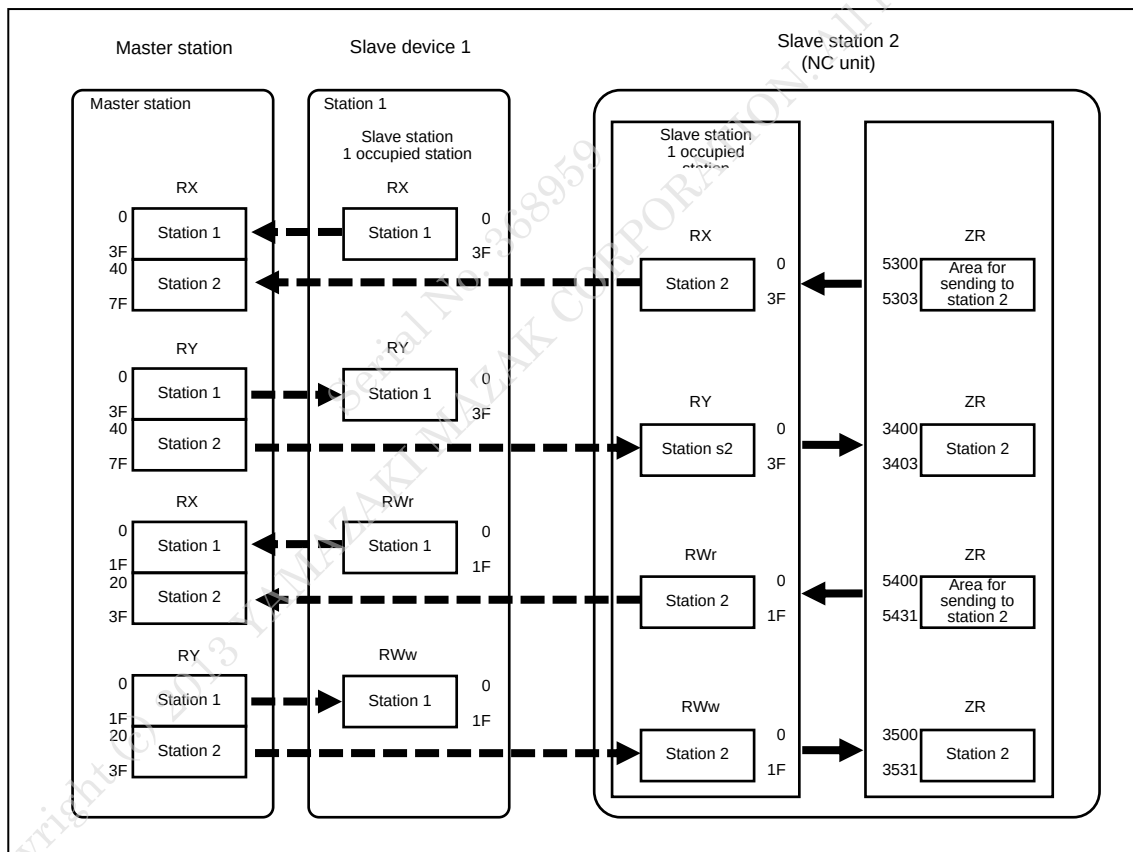
Note : For details of parameters, see “4-1 CC-Link IE Field Basic Parameters” or the separately supplied “PARAMETER LIST/ALARM LIST/M-CODE LIST.”

B. Example of parameter settings with the NC unit set as slave station

Here are some examples of parameter settings when the NC unit serves as the slave station and one master station and one slave station are connected to it.



Shown below are examples in which link devices RX and RY per station are assigned at 64 points, and link devices RWr and RWw per station are assigned at 32 points, based on the assumption that there is only one occupied station in each slave device.



< Examples of settings of the parameters >

1. Enter the operation settings, for example, station type, in parameters **CC1** to **CC5**. It is not necessary to set the master station information in the slave function setting.

Address	Description	Setting
CC1	Communication enabled	1: Enabled
CC2	Setting the type of station	1: Slave station
CC3	Error switchover	0: Reset and stop
CC4	Data link defective station setting	0: Clear
CC5	Output setting during PLC STOP	0: Output

2. Enter the settings in parameters **CS1** to **CS10** for refresh device assigning.

Address	Description	Setting
CS1	Slave function Occupied station	
CS2	Slave function Refresh device project number	1
CS3	Slave function Refresh device name (RX)	ZR
CS4	Slave function Refresh device name (RY)	ZR
CS5	Slave function Refresh device name (RW _r)	ZR
CS6	Slave function Refresh device name (RW _w)	ZR
CS7	Slave function Leading refresh device number (RX)	5300
CS8	Slave function Leading refresh device number (RY)	3400
CS9	Slave function Leading refresh device number (RW _r)	5400
CS10	Slave function Leading refresh device number (RW _w)	3500

Note : For details of parameters, see “4-1 CC-Link IE Field Basic Parameters” or the separately supplied “PARAMETER LIST/ALARM LIST/M-CODE LIST.”

3-4 Communication Error

3-4-1 CC-Link IE Field Basic communication error

1. Error in CC-Link IE Field Basic communication

If an error occurs in CC-Link IE Field Basic communication, communication between the NC unit and the master/slave station will be broken and the alarm **1500 CC-LINK IEF B (M) CONNECT ERR** or **1501 CC-LINK IEF B (S) CONNECT ERR** will be triggered.

If an invalid value is set for the parameter, the alarm **1502 CC-LINK IEF BASIC PARAMETER ERR** will occur.

- If a communication error occurs, the NC unit continues operating, but the alarm is left displayed.
- If a communication error occurs, an error code is displayed in the first argument to the alarm.
For details of error codes, see “3-4-2 Error code list.”

3-4-2 Error code list

An error code displayed in the first argument of the alarm **1500 CC-LINK IEF B (M) CONNECT ERR** or **1501 CC-LINK IEF B (S) CONNECT ERR** is displayed as hexadecimal numbers.

Error code (hexadecimal)	Item	Description
CFC0	Cyclic transmission defect (master station)	Multiple master stations exist in the same network address, so cyclic transmission cannot be implemented. Check to see if another master station exists in the network.
CFC1	Cyclic transmission defect (master station)	A defect occurred during cyclic transmission, preventing cyclic transmission from being resumed. In such a case, switch the power ON again, and carry out data exchange once again.
CFC8	Cyclic transmission defect (master station)	A slave station controlled by another master station exists, so cyclic transmission cannot be implemented. Check to see if another master station exists in the network.
CFD0	Host station defect	Port numbers (61450, 61451) used in CC-Link IE Field Basic are already in use. Carry out a re-examination of ports used by other functions.
CFE0	Cyclic transmission defect (slave station)	Data was transmitted cyclically to a slave station controlled by another master station. In such a case, check to see if another master station is in the network.
CFE1	Cycle transmission defect (slave station)	The number of occupied stations that could not be handled from the master station was designated. Re-examine the setting of the number of stations occupied by the master station.
CFE8	Cycle transmission defect (slave station)	There is no response from a slave station. Check to see if the slave station is connected to the network. Check to see if the slave station disconnect detection settings (timeout time and number of cycles) of the master station are suitable for the slave station.
CFE9	Cycle transmission defect (slave station)	The slave station that has the same IP address as that in the same network exists. Change the IP address of the slave station to a value that is not duplicated.
CFF0	Cycle transmission defect (slave station)	A defect occurred in a slave station. Check the slave station in which the defect occurred.
B001	Host station disconnect	A host station disconnect at a slave station was detected. Check to see if the master station has been connected to the network.
B002	Master station application program stoppage	A stoppage of the application program (PLC function, etc.) of the master station was detected at a slave station. Change the condition of the application program of the master station to an operating condition.
B003	Cyclic transmission stoppage	The slave station received instructions from the master station to stop cyclic transmission. In such a case, restart cyclic transmission at the master station.

The error code displayed as the first argument of alarm **1502 CC-LINK IEF BASIC PARAMETER ERR** is displayed as a 4-digit numerical value.

The first two digits of the 4-digit value indicate classification (letters of the alphabet, such as CE), and the last two digits indicate the number.

Classification	Number
CC	11
CB	12
CS	13
IP address or subnet mask	14

For example, if the first argument is 1105, an error related to the **CC5** parameter has occurred.

Parameter	Item	Description
Erroneous parameter No.	Value is outside parameter setting range	A value or character string outside the setting range of the parameters was set. Re-examine the set value of the displayed parameter number.
CB1	Master function, Number of connected devices	The number of connected devices has not been set. Set the number of connected devices.
CB13	Number of occupied stations of slave device	The number of occupied stations has not been set. Either set the number of occupied stations, or re-examine the number of devices to be connected (parameter CB1).
CB16	Refresh device project number of slave device	A number for a project that cannot be used has been set. Re-examine the number of projects that can be used, and then correct the number.
CB17	Slave device IP address	- The settings of the IP address of the slave device have been duplicated. Re-examine the various settings so that the IP address is not duplicated. - The settings coincide with those of the host station (master station). Repeat the settings so that they do not coincide with the IP address of the host station. - The host station (master station) does not match the network address. Check the IP address of the host station and the subnet mask, and select settings that match those of the network address. Example : Host station IP address: 192.168.200.1 Host station subnet mask: 255.255.255.0 Slave device IP address: 192.168.200.2 to 192.168.200.254 - An IP address has not been set. Set an IP address or re-examine the number of connected units (parameter CB1).
CB22 CB23 CB24 CB25	Leading refresh device number of slave device	- The set PLC device has been set as far as the range exceeding the zone in which the PLC device can be used. - The PLC device coincides with the zone of the PLC device designated by another parameter. Set the leading number of the PLC device while taking into consideration the number of link devices for link refresh.
CS1	Slave function, Number of occupied stations	The number of occupied stations has not been set. Set the number of occupied stations.
CS2	Slave function, Refresh device project number	A number for a project that cannot be used has been set. Re-examine the number of projects that can be used, and then correct the number.
CS7 CS8 CS9 CS10	Slave function, Leading refresh device number	- The set PLC device has been set as far as the range exceeding the zone in which the PLC device can be used. - The PLC device coincides with the zone of the PLC device designated by another parameter. Set the leading number of the PLC device while taking into consideration the number of link devices for refresh.

4 LIST OF PARAMETERS FOR FIELD NETWORK

The parameters used for a field network are enumerated below.

4-1 CC-Link IE Field Basic Parameters

Shown below are the parameters that are used for CC-Link IE Field Basic communications.

1. Common setting parameters of CC-Link IE Field Basic communications

These parameters need to be set for the NC unit to carry out communications over a CC-Link IE Field Basic network. They always need to be set no matter what role the NC unit plays: master or slave station.

Address	Meaning		Description
CC1	Communication enabled		Enables communication to be performed. 0: Disabled 1: Enabled
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0, 1	
CC2	Setting the type of station		Sets the type of station of the NC unit. 0: Master station 1: Slave station
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0, 1	
CC3	Error switchover		Makes a setting which performs a reset stoppage or emits an alarm display, when an error occurs. 0: Reset stoppage 1: Alarm display
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0, 1	
CC4	Data link defective station setting		Specifies whether, when the data link is defective, the input data (RX in the case of the master function, or RY in the case of a slave function) is cleared, or the immediately preceding value is held. 0: Clear 1: Hold
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0, 1	

Address	Meaning		Description
CC5	Output setting during PLC STOP		Specifies whether, when the PLC goes into a STOP condition, the output data (RY in the case of the master function, or RX in the case of a slave function) is held, or cleared. 0: Hold 1: Clear
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0, 1	
CC6 CC16	—		Invalid
	Program type	—	
	Conditions	—	
	Unit	—	
	Setting range	—	

2. CC-Link IE Field Basic master function setting parameters

These parameters are used when the NC unit is used as the master station.

Parameters **CB13** to **CB25** are set for the slave stations connected in the same system as the master station set by **CC2**. They need to be set in the same order as the slave stations are connected.

Address	Meaning		Description
CB1	Master function Number of connected devices		Sets the number of slave devices connected using the master function.
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0 to 4	
CB2	—		Invalid
	Program type	—	
	Conditions	—	
	Unit	—	
	Setting range	—	
CB3	Master function Timeout time		Sets the timeout time for cyclic communication, in units of ms. Note : 1. In the case of 0, the timeout time is deemed to be 100 ms. 2. When a numerical value outside the setting range is set, the alarm 1502 CC-LINK IEF BASIC PARAMETER ERR will occur.
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	ms	
	Setting range	0, 20 to 65535	
CB4 CB6	—		Invalid
	Program type	—	
	Conditions	—	
	Unit	—	
	Setting range	—	

Address	Meaning		Description
CB7	Master function Number of timeout cycles		Sets the number of timeout cycles for cyclic communication. A slave station whose continuous timeover cycle count exceeds the set value is disconnected. Note : 1. In the case of 0, the value is deemed to be 3 times. 2. When a value outside the setting range is set, the alarm 1502 CC-LINK IEF BASIC PARAMETER ERR will occur.
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0, 3, 5, 10	
CB8 CB12	—		Invalid
	Program type	—	
	Conditions	—	
	Unit	—	
	Setting range	—	
CB13	Number of occupied stations of slave device		Sets the number of occupied stations of the slave device. Up to four stations can be occupied by one slave device. Example : If the first device occupies four stations, the second device will start from Station 5. • 1st device: 4 stations occupied (Station 1 to 4) • 2nd device: 1 station occupied (Station 5)
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0 to 4	
CB14	—		Invalid
	Program type	—	
	Conditions	—	
	Unit	—	
	Setting range	—	

Address	Meaning		Description
CB15	Reservation station setting of slave device		Sets the slave device to the reservation station 0: No setting 1: Reservation station Note : 1. When a station is set to the reservation station, communication with the slave device does not take place. 2. By setting a station to a reservation station, you can reserve a scheduled station number for future use. Example : In the case where the first device is a reservation station that occupies four stations, the second device will start from Station 5. Communication will not take place in the case of Stations 1 to 4. • 1st device (reservation station): 4 stations (Station 1 to 4) occupied (Communication does not take place.) • 2nd device: 1 station (Station 5) occupied
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0, 1	
CB16	Refresh device project number of slave device		Sets the project number of the PLC device that performs link refresh with the link device of the slave device.
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0 to 6	
CB17	Slave device IP address		Sets the IP address of the slave device. Note : 1. Set the IP address of the slave device to the same network as that of the master station. 2. You cannot communicate with a device belonging to a different network.
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0.0.0.0 to 255.255.255.255	
CB18	Refresh device name (RX) of slave device		Sets the device name of the PLC device which performs link refresh with the link device "RX" of the slave device. Remark : Device that can perform link refresh: ZR
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	— (invalid), ZR	

Address	Meaning		Description
CB19	Refresh device name (RY) of slave device		Sets the device name of the PLC device which performs link refresh with the link device "RY" of the slave device. Remark : Device that can perform link refresh: ZR
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	— (invalid), ZR	
CB20	Refresh device name (RW _r) of slave device		Sets the device name of the PLC device which performs link refresh with the link device "RW _r " of the slave device. Remark : Device that can perform link refresh: ZR
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	— (invalid), ZR	
CB21	Refresh device name (RW _w) of slave device		Sets the device name of the PLC device which performs link refresh with the link device "RW _w " of the slave device. Remark : Device that can perform link refresh: ZR
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	— (invalid), ZR	
CB22	Leading refresh device number (RX) of slave device		Sets the leading device number of the ZR register which performs link refresh with the slave device link device "RX." Note : If 0 is set for this parameter, the alarm 1502 CC-LINK IEF BASIC PARAMETER ERR will occur.
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0, 3400 to 3975	

Address	Meaning		Description
CB23	Leading refresh device number (RY) of slave device		Sets the leading device number of the ZR register which performs link refresh with the slave device link device "RY." Note : If 0 is set for this parameter, the alarm 1502 CC-LINK IEF BASIC PARAMETER ERR will occur.
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0, 5300 to 5875	
CB24	Leading refresh device number (RW _r) of slave device		Sets the leading device number of the ZR register which performs link refresh with the slave device link device "RW _r ." Note : If 0 is set for this parameter, the alarm 1502 CC-LINK IEF BASIC PARAMETER ERR will occur.
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0, 3400 to 3975	
CB25	Leading refresh device number (RW _w) of slave device		Sets the leading device number of the ZR register which performs link refresh with the slave device link device "RW _w ." Note : If 0 is set for this parameter, the alarm 1502 CC-LINK IEF BASIC PARAMETER ERR will occur.
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0, 5300 to 5875	

3. CC-Link IE Field Basic slave function setting parameters

Set these parameters when the NC unit is used as a slave station.

Address	Meaning		Description
CS1	Slave function Occupied station		Sets the number of occupied stations of the slave function. Up to 4 stations can be occupied.
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0 to 4	
CS2	Slave function Refresh device project number		Sets the project number of the PLC device that performs link refresh with the link device of the slave function.
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0 to 6	
CS3	Slave function Refresh device name (RX)		Sets the device name of the PLC device which performs link refresh with the link device "RX" of the slave function. Remark : Device that can perform link refresh: ZR
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	— (invalid), ZR	
CS4	Slave function Refresh device name (RY)		Sets the device name of the PLC device which performs link refresh with the link device "RY" of the slave function. Remark : Device that can perform link refresh: ZR
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	— (invalid), ZR	

Address	Meaning		Description
CS5	Slave function Refresh device name (RWr)		Sets the device name of the PLC device which performs link refresh with the link device "RWr" of the slave function. Remark : Device that can perform link refresh: ZR
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	— (invalid), ZR	
CS6	Slave function Refresh device name (RWw)		Sets the device name of the PLC device which performs link refresh with the link device "RWw" of the slave function. Remark : Device that can perform link refresh: ZR
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	— (invalid), ZR	
CS7	Slave function Leading refresh device number (RX)		Sets the leading device number of the ZR register which performs link refresh with the slave function link device "RX." Note : If 0 is set for this parameter, the alarm 1502 CC-LINK IEF BASIC PARAMETER ERR will occur.
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0, 5300 to 5875	
CS8	Slave function Leading refresh device number (RY)		Sets the leading device number of the ZR register which performs link refresh with the slave function link device "RY." Note : If 0 is set for this parameter, the alarm 1502 CC-LINK IEF BASIC PARAMETER ERR will occur.
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0, 3400 to 3975	

Address	Meaning		Description
CS9	Slave function Leading refresh device number (RW _r)		Sets the leading device number of the ZR register which performs link refresh with the slave function link device "RW _r ." Note : If 0 is set for this parameter, the alarm 1502 CC-LINK IEF BASIC PARAMETER ERR will occur.
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0, 5300 to 5875	
CS10	Slave function Leading refresh device number (RW _w)		Sets the leading device number of the ZR register which performs link refresh with the slave function link device "RW _w ." Note : If 0 is set for this parameter, the alarm 1502 CC-LINK IEF BASIC PARAMETER ERR will occur.
	Program type	M, E	
	Conditions	Power is turned ON.	
	Unit	—	
	Setting range	0, 3400 to 3975	
CS11 CS15	—		Invalid
	Program type	—	
	Conditions	—	
	Unit	—	
	Setting range	—	